Daily Literature Review — 2026-11-05

Daily Literature Review: AI and machine learning in protein function prediction

Date: 2026-11-05

Research Question

How are machine-learning methods being used to predict protein function, what datasets and evaluation metrics are used, and what limitations remain?

Summary

This document compiles 20 recent papers related to the topic. AI drafting was unavailable, so the content below is built directly from bibliographic metadata. Each entry includes title, authors, year, source, DOI/URL, and a reminder to extract findings from the original paper. This is an evidence-collection scaffold, not a finished scientific review.

Evidence Table (Bibliographic)

2026

Rapid directed evolution guided by protein language models and epistatic interactions.

- Authors: Tran VQ, Nemeth M, Bartie LJ, Chandrasekaran SS, Fanton A, Moon HC, Hie BL, Konermann S

- Source: PubMed

- PMID: 41712694

- Link: https://pubmed.ncbi.nlm.nih.gov/41712694/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Decoding and deciphering a subcellular ZIP code system.

- Authors: Ille AM, Markosian C, Pasqualini R, Arap W

- Source: PubMed

- PMID: 41820158

- Link: https://pubmed.ncbi.nlm.nih.gov/41820158/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Discovery of natural RORγt inhibitor using machine learning, virtual screening, and in vivo validation.

- Authors: Yoo H, Han SJ, Lee JE, Cho C, Hong D, Jeong B, Kim S, Jung GY

- Source: PubMed

- PMID: 40915561

- Link: https://pubmed.ncbi.nlm.nih.gov/40915561/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Systematic Multi-Target QSAR Benchmarking: Machine Learning Algorithms, Molecular Descriptors, and Validation.

- Authors: Alshehade SA, Al Jabal G, Jebril IH

- Source: PubMed

- PMID: 42300227

- Link: https://pubmed.ncbi.nlm.nih.gov/42300227/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2025

LCN2 drives ferroptosis-associated ischemia-reperfusion injury after renal transplantation: integrated machine learning and in vivo validation.

- Authors: Wu Z, Yu B, He Q, Huang C

- Source: PubMed

- PMID: 41205031

- Link: https://pubmed.ncbi.nlm.nih.gov/41205031/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Can AI-Predicted Complexes Teach Machine Learning to Compute Drug Binding Affinity?

- Authors: Hsu WT, Grevtsev S, Herz AM, Douglas T, Magarkar A, Biggin PC

- Source: PubMed

- PMID: 41369266

- Link: https://pubmed.ncbi.nlm.nih.gov/41369266/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Deciphering the mechanism of baicalein in cervical cancer via bioinformatics, machine learning and computational simulations: PIM1 and CDK2 are key target proteins.

- Authors: Wang S, Liu C, Ye D, Qi J, Xing Y, Wang J, Fan X, Li X

- Source: PubMed

- PMID: 40339869

- Link: https://pubmed.ncbi.nlm.nih.gov/40339869/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning Prediction of 90-Day Mortality in HBV-Related ACLF Using Olink-Derived Inflammatory Protein Signatures.

- Authors: Zhang Y, Sun L, Liu H, Shi K, Lu Y, Feng Y, Wang X

- Source: PubMed

- PMID: 41255250

- Link: https://pubmed.ncbi.nlm.nih.gov/41255250/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning driven prediction of drug efficacy in lung cancer: based on protein biomarkers and clinical features.

- Authors: Li J, Chen A, Liu Z, Wei S, Zhang J, Chen J, Shi C

- Source: PubMed

- PMID: 40355026

- Link: https://pubmed.ncbi.nlm.nih.gov/40355026/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Applications of Machine Learning Approaches for the Discovery of SARS-CoV-2 PLpro Inhibitors.

- Authors: Pal S, Nance KD, Joshi DR, Kales SC, Ye L, Hu X, Shamim K, Zakharov AV

- Source: PubMed

- PMID: 39818814

- Link: https://pubmed.ncbi.nlm.nih.gov/39818814/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2024

AI-Driven Deep Learning Techniques in Protein Structure Prediction.

- Authors: Chen L, Li Q, Nasif KFA, Xie Y, Deng B, Niu S, Pouriyeh S, Dai Z

- Source: PubMed

- PMID: 39125995

- Link: https://pubmed.ncbi.nlm.nih.gov/39125995/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning-based reproducible prediction of type 2 diabetes subtypes.

- Authors: Tanabe H, Sato M, Miyake A, Shimajiri Y, Ojima T, Narita A, Saito H, Tanaka K

- Source: PubMed

- PMID: 39168869

- Link: https://pubmed.ncbi.nlm.nih.gov/39168869/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning interpretable models of cell mechanics from protein images.

- Authors: Schmitt MS, Colen J, Sala S, Devany J, Seetharaman S, Caillier A, Gardel ML, Oakes PW

- Source: PubMed

- PMID: 38194965

- Link: https://pubmed.ncbi.nlm.nih.gov/38194965/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AKT and EZH2 inhibitors kill TNBCs by hijacking mechanisms of involution.

- Authors: Schade AE, Perurena N, Yang Y, Rodriguez CL, Krishnan A, Gardner A, Loi P, Xu Y

- Source: PubMed

- PMID: 39385030

- Link: https://pubmed.ncbi.nlm.nih.gov/39385030/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Multi-Omics Integration With Machine Learning Identified Early Diabetic Retinopathy, Diabetic Macula Edema and Anti-VEGF Treatment Response.

- Authors: Pang Y, Luo C, Zhang Q, Zhang X, Liao N, Ji Y, Mi L, Gan Y

- Source: PubMed

- PMID: 39671223

- Link: https://pubmed.ncbi.nlm.nih.gov/39671223/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Enhancing the Predictive Power of Machine Learning Models through a Chemical Space Complementary DEL Screening Strategy.

- Authors: Suo Y, Qian X, Xiong Z, Liu X, Wang C, Mu B, Wu X, Lu W

- Source: PubMed

- PMID: 39441849

- Link: https://pubmed.ncbi.nlm.nih.gov/39441849/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2023

Novel machine learning approaches revolutionize protein knowledge.

- Authors: Bordin N, Dallago C, Heinzinger M, Kim S, Littmann M, Rauer C, Steinegger M, Rost B

- Source: PubMed

- PMID: 36504138

- Link: https://pubmed.ncbi.nlm.nih.gov/36504138/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Hierarchical graph learning for protein-protein interaction.

- Authors: Gao Z, Jiang C, Zhang J, Jiang X, Li L, Zhao P, Yang H, Huang Y

- Source: PubMed

- PMID: 36841846

- Link: https://pubmed.ncbi.nlm.nih.gov/36841846/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2022

Machine Learning Approaches for Metalloproteins.

- Authors: Yu Y, Wang R, Teo RD

- Source: PubMed

- PMID: 35209064

- Link: https://pubmed.ncbi.nlm.nih.gov/35209064/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AI-Based Protein Structure Prediction in Drug Discovery: Impacts and Challenges.

- Authors: Schauperl M, Denny RA

- Source: PubMed

- PMID: 35727311

- Link: https://pubmed.ncbi.nlm.nih.gov/35727311/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Recommended Next Steps

1. Open the attached individual PDFs (or follow the DOI/URL links).

2. For each relevant paper, extract: main method, datasets used, evaluation metrics, key quantitative results, and stated limitations.

3. Record contradictions or gaps across papers.

4. Only after verification, move evidence into a formal manuscript.

Scientific Rule

AI is an assistant, not a source. Never submit an AI-generated or placeholder claim without checking the original paper.

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